LAB-ASSIGNMENT : 2

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Objective:

• To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

```
mysql> CREATE DATABASE DBMS_LAB2;
Query OK, 1 row affected (0.10 sec)

mysql> USE DBMS_LAB2;
Database changed
```

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).
The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

- How do you create a table named CLASS with appropriate data types?

```
mysql> CREATE TABLE CLASS (
    ->     ID INT PRIMARY KEY,
    ->     Name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.16 sec)
```

```
mysql> INSERT INTO CLASS VALUES (1,'Bravo');
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO CLASS VALUES (2,'Alex');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO CLASS VALUES (4,'Cheng');
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

- How do you create a table named EMPLOYEES with the specified attributes?

```
mysql> CREATE TABLE EMPLOYEES (
    ->     Employee_Id INT PRIMARY KEY,
    ->     First_Name VARCHAR(50),
    ->     Last_Name VARCHAR(50),
    ->     Email VARCHAR(100),
    ->     Phone_Number VARCHAR(20),
    ->     Hire_Date DATE,
    ->     Job_Id VARCHAR(20),
    ->     Salary INT,
    ->     Commission_Pct DECIMAL(5,2),
    ->     Manager_Id INT,
    ->     Department_Id INT
    -> );
Query OK, 0 rows affected (0.04 sec)
```

- How do you create a table named Client_master with the specified attributes?

```
mysql> CREATE TABLE Client_master (
    ->     ClientNO VARCHAR(10) PRIMARY KEY,
    ->     Name VARCHAR(50),
    ->     Address VARCHAR(100),
    ->     City VARCHAR(50),
    ->     State VARCHAR(50),
    ->     Pincode INT,
    ->     bal_due INT
    -> );
Query OK, 0 rows affected (0.04 sec)
```

- What SQL statement would you use to rename the table Client_master to Client12?

```
mysql> RENAME TABLE Client_master TO Client12;
Query OK, 0 rows affected (0.04 sec)
```

- How do you display the bal_due column heading as "BALANCE"?

```
mysql> SELECT bal_due AS BALANCE FROM Client12;
Empty set (0.01 sec)
```

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- INSERT INTO class VALUES (5,'Rahul');
- COMMIT;

```
mysql> INSERT INTO class VALUES (5,'Rahul');
Query OK, 1 row affected (0.07 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from class;
+----+-------+
| ID | Name  |
+----+-------+
|  1 | Bravo |
|  2 | Alex  |
|  4 | Cheng |
|  5 | Rahul |
+----+-------+
4 rows in set (0.04 sec)
```

- UPDATE class SET name = 'Abhijeet' WHERE id='5';
- SAVEPOINT A;

```
mysql> UPDATE class SET name='Abhijeet' WHERE id=5;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SAVEPOINT A;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from class;
+----+----------+
| ID | Name     |
+----+----------+
|  1 | Bravo    |
|  2 | Alex     |
|  4 | Cheng    |
|  5 | Abhijeet |
+----+----------+
4 rows in set (0.00 sec)
```

- INSERT INTO class VALUES (6,'Chris');
- SAVEPOINT B;

```
mysql> INSERT INTO class VALUES (6,'Chris');
Query OK, 1 row affected (0.00 sec)

mysql> SAVEPOINT B;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from class;
+----+----------+
| ID | Name     |
+----+----------+
|  1 | Bravo    |
|  2 | Alex     |
|  4 | Cheng    |
|  5 | Abhijeet |
|  6 | Chris    |
+----+----------+
5 rows in set (0.00 sec)
```

- INSERT INTO class VALUES (7,’Bravo’);
- SAVEPOINT C;
- SELECT * FROM class;

```
mysql> INSERT INTO class VALUES (7,'Bravo');
Query OK, 1 row affected (0.00 sec)

mysql> SAVEPOINT C;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from class;
+----+----------+
| ID | Name     |
+----+----------+
|  1 | Bravo    |
|  2 | Alex     |
|  4 | Cheng    |
|  5 | Abhijeet |
|  6 | Chris    |
|  7 | Bravo    |
+----+----------+
6 rows in set (0.00 sec)
```

- ROLLBACK TO B;
- SELECT * FROM class;
- ROLLBACK TO A;

```
mysql> rollback to b;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from class;
+----+----------+
| ID | Name     |
+----+----------+
|  1 | Bravo    |
|  2 | Alex     |
|  4 | Cheng    |
|  5 | Abhijeet |
|  6 | Chris    |
+----+----------+
5 rows in set (0.01 sec)

mysql> rollback to a;
Query OK, 0 rows affected (0.00 sec)
```

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.

```
mysql> SELECT Employee_Id, First_Name, Last_Name, Salary
    -> FROM EMPLOYEES;
+-------------+------------+-----------+--------+
| Employee_Id | First_Name | Last_Name | Salary |
+-------------+------------+-----------+--------+
|         101 | John       | Austin    |   5000 |
|         102 | David      | King      |   4500 |
|         103 | Sara       | Austin    |   6000 |
|         104 | Neha       | Sharma    |   4800 |
|         105 | Raj        | Patel     |   5200 |
+-------------+------------+-----------+--------+
5 rows in set (0.02 sec)
```

2. List out the employees who work under manager 100.

```
mysql> SELECT First_Name, Last_Name
    -> FROM EMPLOYEES
    -> WHERE Manager_Id=100;
+------------+-----------+
| First_Name | Last_Name |
+------------+-----------+
| John       | Austin    |
| David      | King      |
| Neha       | Sharma    |
+------------+-----------+
3 rows in set (0.05 sec)
```

3. Find the names of the employees who have a salary greater than or equal to 4800.

```
mysql> SELECT First_Name, Last_Name
    -> FROM EMPLOYEES
    -> WHERE Salary>=4800;
+------------+-----------+
| First_Name | Last_Name |
+------------+-----------+
| John       | Austin    |
| Sara       | Austin    |
| Neha       | Sharma    |
| Raj        | Patel     |
+------------+-----------+
4 rows in set (0.00 sec)
```

4. List out the employees whose last name is ‘AUSTIN’.

```
mysql> SELECT First_Name
    -> FROM EMPLOYEES
    -> WHERE Last_Name='Austin';
+------------+
| First_Name |
+------------+
| John       |
| Sara       |
+------------+
2 rows in set (0.02 sec)
```

5. Find the names of the employees who work in departments 60, 70, and 80.

```
mysql> SELECT First_Name
    -> FROM EMPLOYEES
    -> WHERE Department_Id IN (60,70,80);
+------------+
| First_Name |
+------------+
| John       |
| David      |
| Sara       |
| Neha       |
| Raj        |
+------------+
5 rows in set (0.00 sec)
```

6. Display the unique Manager_Id.

```
mysql> SELECT DISTINCT Manager_Id
    -> FROM EMPLOYEES;
+------------+
| Manager_Id |
+------------+
|        100 |
|        101 |
+------------+
2 rows in set (0.02 sec)
```

Part C: Updating Records (Client_master Table)

- Insert five records into the Client_master table.

```
mysql> INSERT INTO Client12 VALUES
    -> ('C123','Aman','MG Road','Delhi','Delhi',110001,6000),
    -> ('C124','Bhavna','Park Street','Mumbai','Maharashtra',400001,4500),
    -> ('C125','Chirag','Brigade Road','Bangalore','Karnataka',560001,7000),
    -> ('C126','Diya','Anna Salai','Chennai','Tamil Nadu',600001,3000),
    -> ('C127','Eshan','Banjara Hills','Hyderabad','Telangana',500001,8000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

- Find the names of clients whose bal_due > 5000.

```
mysql> SELECT Name
    -> FROM Client12
    -> WHERE bal_due>5000;
+--------+
| Name   |
+--------+
| Aman   |
| Chirag |
| Eshan  |
+--------+
3 rows in set (0.00 sec)
```

- Change the bal_due of ClientNO "C123" to Rs. 5100.

```
mysql> UPDATE Client12
    -> SET bal_due=5100
    -> WHERE ClientNO='C123';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from client12;
+----------+--------+---------------+-----------+-------------+---------+---------+
| ClientNO | Name   | Address       | City      | State       | Pincode | bal_due |
+----------+--------+---------------+-----------+-------------+---------+---------+
| C123     | Aman   | MG Road       | Delhi     | Delhi       |  110001 |    5100 |
| C124     | Bhavna | Park Street   | Mumbai    | Maharashtra |  400001 |    4500 |
| C125     | Chirag | Brigade Road  | Bangalore | Karnataka   |  560001 |    7000 |
| C126     | Diya   | Anna Salai    | Chennai   | Tamil Nadu  |  600001 |    3000 |
| C127     | Eshan  | Banjara Hills | Hyderabad | Telangana   |  500001 |    8000 |
+----------+--------+---------------+-----------+-------------+---------+---------+
5 rows in set (0.07 sec)
```